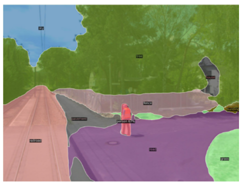
[LinkedIn](#)
wasilmohd1@gmail.com

Computer Vision/Neural Rendering Portfolio - Wasil

(3D) Computer Vision Tasks: Object Classification, Object Detection, Segmentation (Instance/Semantic/Panoptic), Depth Estimation, CNN-based Regression, Conditional GANs, Neural Rendering, Structure-from-Motion (SfM), 3D Scene Reconstruction, Mesh Generation, Point Cloud Processing, Simulation

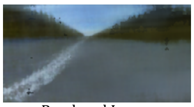
Segmentation / Depth Estimation



- Panoptic Segmentation: Fine-tuned **PanopticFCN** for pedestrian segmentation for ADAS/Autonomous Trains.
- Trained depth estimation models with consistency-aware losses and train/test-time augmentations to stress-test robustness.

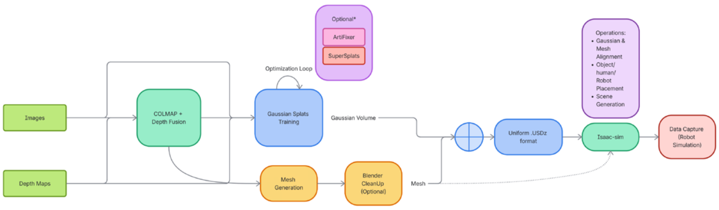
Neural Rendering

Original Image

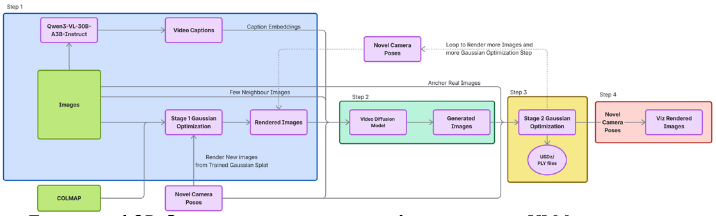


Rendered Image

- Developed RGB/LiDAR Neural Rendering pipelines, using COLMAP for camera poses, Point Cloud Registration, Mesh Generation (Marching Cubes, Poisson, Nvblox), training NeRFs/Gaussian Splats.
- Simulated robotic motion through 3D Gaussian volumes and extracted meshes, automating multi-sensor RGB-D data collection along dynamic paths.
- Implemented Tiny NeRF for forward rendering of road scenes; managed camera extrinsics.

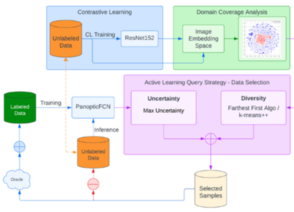


Fine-tuning Gaussian Volumes to Fill Blind Spots



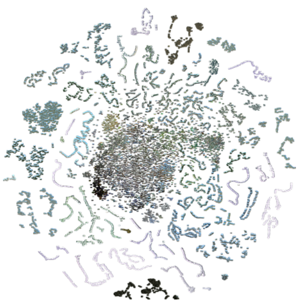
- Fine-tuned 3D Gaussian representations by generating VLM scene captions and camera trajectories for novel view synthesis. Used video diffusion models to inpaint under-observed blind spots, distilling synthetic frames back into a second-stage Gaussian optimization to eliminate volumetric artifacts.

Active Learning



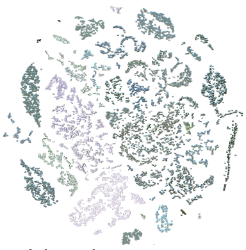
- Performed Active Learning for Panoptic Segmentation of pedestrians.
- Scaled test cases every cycle to maintain consistent train-test data distributions.
- Conducted evaluation on unseen test data in every cycle.

Visualizing Image Embedding space for a sequence of images



- Extracted high-dimensional ResNet embeddings to visualize image sequences via t-SNE.
- Applied clustering (DBSCAN, OPTICS, Agglomerative) to automate Scenario Mining.
- Scenario Mining across urban cities, indoor/outdoor depots, countryside, forests, and open fields, identifying edge cases, such as pedestrians in tunnels and diverse pedestrian activities under different lighting.

Contrastive Learning



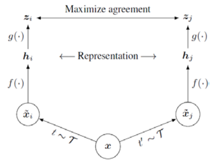
Newly learned Representation Space

- Trained a Siamese ResNet-152 backbone using NT-Xent loss to learn robust image representations.
- Mapped feature space to cluster semantically similar images while distancing dissimilar samples.
- Used learned embeddings for downstream tasks, including diverse data sampling.

Uncertainty Estimation

- Standard Strategies: Margin Sampling, Least Confidence, Entropy Criterion.
- Studied Adversarial Attacks for uncertainty-based sample selection.
- Explored Methods: Query-by-Committee (QBC), Monte-Carlo (MC) Dropout, Test-Time Augmentation (TTA), Ensemble Methods.
- QBC with TTA workflow utilizing KL-Divergence to quantify distances between predictive probability distributions.

Diversity Query Strategy



- Fine-tuned a high-dimensional Embedding space with **Contrastive Learning**
- High-dimensional features are mapped to 2D t-SNE space.
- Diversity Sampling: Used Farthest First/Gonzalez Algorithm for querying samples.

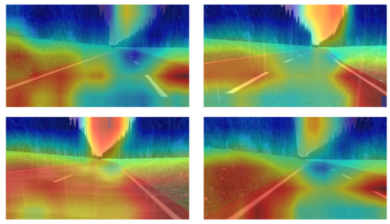
(conditional) GANs

- Developed GANs and conditional GANs (cGANs) from scratch
- Utilized CelebA dataset to generate images with specific attribute combinations.

CNN-based Regression for Autonomous Driving

- **Self-Driving Car**: Trained a 5-layer CNN architecture to predict steering angles and perform road-line detection.
- Control Logic: Used predicted steering angles to determine vehicle velocity.

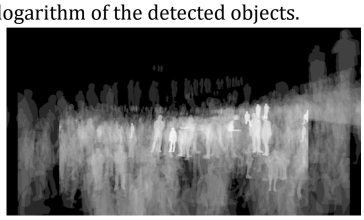
Grad-CAM Analysis



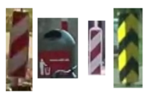
- Performed Grad-CAM analysis to identify pixels driving steering angle predictions.
- Analyzed model behavior in Out-of-Domain scenarios such as rain, fog, and night sky.
- Evaluated model reactions under domain shift.

Image/3D Point Cloud Processing

- Analyzed objects based on location and shape utilizing heatmaps.
- Calculated heatmaps by taking the logarithm of the detected objects.



- Filtered False Positives from inference results before passing to domain experts for labeling.
- Used pixel-wise distributions and Euclidean distance to extract top-3 matching and non-matching samples per image for Contrastive Loss.
- Verified 3D point cloud accuracy by performing Euclidean distance benchmarks.
- Managed 3D data pipelines by converting COLMAP binary (.bin) outputs to .ply and text formats for cross-tool processing.



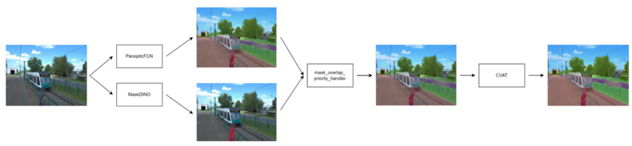
False Positives

Data Curation

- Semantic Retrieval: Performed data curation for specific scenario retrieval across massive unlabeled datasets.
- Used Zero-shot Transfer for all data curation and retrieval tasks.
- Vision Transformers: Used BLIP/BLIP-2 for image-text semantic retrieval.

Semi-Automatic Annotation and Data Labeling Policy

- Automate the Annotation pipeline for faster processing and training.
- Integrated SAM Docker within CVAT for native AI-assisted labeling.
- Used YOLO, SAM, MaskDINO, Mask2Former, and PanopticFCN for semi-automatic annotation.



- Defined class taxonomy and labeling guidelines, addressed edge cases for consistency.
- Handled RLE to Polygon format conversion and vice-versa.
- Conducted annotation sessions using CVAT.

Evaluation Metrics

- Average Precision (COCO) and Pedestrian Detection Rate
- Inception score (IS) and Fréchet Inception Distance
- PSNR
- Qualitative Analysis
- In-Domain vs Out-of-Domain image evaluation
- Spatial Analysis: Bounding box height vs Width ratio, Segmentation mask area count.

In-Domain Image Sample

Out-of-Domain Image Sample